

PRODUCT REQUIREMENTS DOCUMENT (PRD)

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Product	Campus connect
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1. Executive Summary (1-minute read)

1.1 Background

University students lose valuable academic opportunities because forming effective study groups across sections is manual, fragmented across multiple platforms, and largely dependent on chance. Campus Connect addresses this by launching an AI-powered study group matchmaker that connects students within the same course based on complementary strengths, weaknesses, and availability, enabling faster and more effective collaboration. By validating academic matching as our core MVP, the platform establishes the foundation for a scalable campus connectivity system that can later expand into senior mentorship and cross-sectional collaboration initiatives.

1.2 Problem Statement

Problem:During exam preparation, students struggle to form effective study groups because the process is fragmented, manual, and unpredictable. Group formation requests are scattered across multiple platforms, requiring repeated outreach and schedule coordination during already high-stress periods. Even when groups are formed, students have little visibility into peer compatibility, often resulting in mismatched preparation levels or unclear commitment.

Why it matters: When a study group formation depends on chance and manual coordination, students waste valuable exam preparation time and experience unnecessary stress. Many either settle for ineffective groups formed out of convenience or choose to study alone despite benefiting from collaboration. This reduces learning efficiency, weakens peer engagement, and limits academic performance during high-stakes exam periods.

1.3 Proposed Solution (High-level)

- Campus Connect solves fragmented peer coordination through an AI-driven matchmaking system that pairs students based on complementary academic profiles and aligned availability.
- The platform transforms study group formation from a time-consuming manual process into an instant, data-driven recommendation—delivering the right study partners at the right time eliminating discovery friction and enabling them to focus on actual learning rather than logistics.

2. Solution Overview(MVP)

2.1 Minimum Viable Product (MVP)

MVP Definition: Course-Specific AI Study Group Matchmaker with Peer-Complement Intelligence

The MVP delivers a focused matching system for a single course section, validating whether students will trust and act on AI-generated study group recommendations based on complementary academic strengths.

How It Works:

Input Phase (Profile Creation - 3 minutes per course):

Students complete a streamlined profile for their enrolled course (e.g., "CS201 - Data Structures, Section B")

1) Self-Assessment

Rate confidence level across major course topics through a series of questions.

Example: "Strong in Arrays & Sorting, Weak in Graph Algorithms & Dynamic Programming"

Also add a Learning Role Preference:

"I can help others with: [Topics]"

"I need help with: [Topics]"

2) Study Goals, Availability and study style

Select primary objective (Exam Preparation / Assignment Help / Concept Clarity / All)

Indicate preferred study times (weekday evenings, weekend mornings, etc.)

Brief preferences (problem-solving focus, theory review, practice tests, teaching-based learning)

3) AI Processing (Peer-Complement Matching Algorithm):

The system analyzes all students enrolled in the same course section to create balanced study groups (3-5 members) using:

4) Complementary Strength Matching

Pair students strong in Topic A with those needing help in Topic A, and vice versa—creating mutual teaching opportunities rather than same-skill clustering

Ensure at least 2 overlapping time slots per week for the group

Goal Compatibility: Prioritize grouping students with similar study objectives (e.g., all focused on exam prep vs. assignment collaboration)

Balanced Group Composition: Avoid groups with all "struggling" or all "confident" students; aim for mixed-capability groups where peer teaching can occur

5) Output & Group Formation:

What students receive:

1. Primary Recommendation: One curated 3-5 person study group with:

Member names and brief academic profiles

Group compatibility score (0-100%)

2. Explanation: "This group balances your graph algorithm weakness with Priya's strength, while you can help others with sorting concepts"

Shared availability windows

3. Action Options:

Accept group assignment (express interest)

Request alternative matches (if available)

4. Group Activation: When 3+ members accept, the group is formed and contact details are shared. Students coordinate independently via their preferred platform (WhatsApp, Telegram, etc.)

Validation Focus:

This MVP tests **three critical assumptions**:

Will students accept algorithmic matches over self-selected friends?

Do peer-complement groups (mixed strengths) perform better than random/same-skill groups?

Will time-pressed students invest 3 minutes per course to enable matching?

Scope Constraints:

(avoid these to categorize them into couldbes' and won'tbes')

- In-app messaging or chat features (groups use existing platforms)
- Calendar/scheduling integration tools
- Campus-wide social feed or announcements
- Cross-course or multi-course matching
- Senior mentorship or placement guidance
- Cultural event coordination
- Multi-university or multi-section rollout (single section pilot only)

Pilot Scope:

Target: Single course section (e.g., CS201 Data Structures - Section B, Second Year)

User Base: 40-60 students in the pilot section

Timeline: Launch 3 weeks before mid-term exams (highest urgency period)

Success Threshold: 60% of active users successfully matched into groups that meet at least twice

Scalability Path:

While the MVP is deliberately constrained to one course section, the architecture is designed for future expansion:

Phase 2: Multi-section matching within the same course

Phase 3: Department-wide deployment across multiple courses

Phase 4: Cross-year mentorship features (seniors helping juniors)

Phase 5: Broader connectivity (placement guidance, cultural events, cross-departmental collaboration)

The peer-complement matching algorithm developed for academic groups becomes the foundation for all future matching use cases—whether connecting students with senior mentors or forming teams for hackathons.

3. USER STORIES & ACCEPTANCE CRITERIA

User Story 1: Quick Profile Creation

As a second-year student preparing for exams,

I want to create my academic profile in under 5 minutes,

So that I can quickly access study group recommendations without wasting valuable study time.

Acceptance Criteria:

1. Profile creation requires maximum 3 screens with auto-save functionality
2. User receives confirmation message immediately upon successful submission

3. No text-heavy inputs; all selections use dropdowns/sliders for speed

User Story 2: Intelligent Peer-Complement Matching

As a CS student strong in algorithms but weak in data structures,
I want to be matched with peers who have complementary strengths,
So that we can teach each other and fill knowledge gaps effectively.

Acceptance Criteria:

1. Algorithm pairs students where A's strengths match B's weaknesses (minimum 2 complementary topics)
2. Group recommendation displays clear rationale explaining the complementary match
3. Compatibility score reflects topic complementarity as 40% of total calculation

User Story 3: Schedule-Compatible Group Formation

As a student with limited free time between classes and work,
I want to be matched only with students who share my available study times,
So that coordinating meetings doesn't become another scheduling burden.

Acceptance Criteria:

1. Algorithm only forms groups where all members share minimum 2 overlapping time slots
2. Group recommendation highlights shared availability windows clearly
3. If no schedule-compatible matches exist, user receives notification to try again later

User Story 4: Transparent Match Recommendations

As a student skeptical about AI recommendations,
I want to understand why I was matched with specific peers,
So that I can trust the system's judgment before accepting.

Acceptance Criteria:

1. Each recommendation shows compatibility score (0-100%), rationale, and shared availability
2. User can view basic profiles of recommended members before accepting
3. If declined, system offers one alternative match with different explanation

User Story 5: Mutual Consent Group Formation

As a student concerned about forced group assignments,
I want to explicitly accept or decline my group,
So that I only join groups I'm genuinely committed to.

Acceptance Criteria:

1. No automatic enrollment; users must click "Accept Group" to confirm participation
2. Group activates only when minimum 3 members accept within 48 hours
3. Users see real-time acceptance status while waiting for other members

User Story 6: Seamless Contact Exchange

As a student whose group has been confirmed,
I want to immediately receive contact information for all members,
So that I can quickly schedule our first study session.

Acceptance Criteria:

1. Upon activation, all members receive email with names, emails, and phone numbers within 5 minutes
2. In-platform dashboard displays confirmed group member contact details
3. Contact exchange occurs only after mutual acceptance

4. Goals & Success Metrics

4.1 Goals

Primary Goal:

Validate that AI-driven peer-complement matching can successfully form effective study groups faster and with higher satisfaction than manual, self-selection methods.

Secondary Goals:

- Demonstrate that algorithmically-matched groups show higher engagement and persistence than randomly formed groups
- Gather user feedback and behavioral data to refine the matching algorithm for future iterations
- Achieve meaningful adoption within the pilot course section during the exam preparation period

4.2 Success Metrics (KPIs)

Differentiating them through the AARRR framework funnel.

→ ACQUISITION - Getting users to sign up

Metric	Current Baseline	Target	Notes
Total sign-ups from pilot section	0 (new product)	35-40 students (60-70% of section)	Launch 3 weeks before exams to maximize urgency
Sign-up conversion rate	N/A	65% of students who view landing page	Measure effectiveness of initial outreach
Time to first sign-up after launch	N/A	Within 48 hours	Indicates market pull and messaging clarity

→ ACTIVATION - Users complete meaningful first action

Metric	Current Baseline	Target	Notes
Profile completion rate	0%	75% of sign-ups	% who complete full course profile (topics, availability, preferences)
Average time to complete profile	N/A	< 4 minutes	Validates friction-free onboarding
Topic assessment completion	0%	80% of profiles	% who rate confidence across all major course topics
Students who receive their first match recommendation	0%	70% of completed profiles	Indicates sufficient user pool for matching

→ RETENTION - Users return and continue using the product

Metric	Current Baseline	Target	Notes
Students who accept group assignment	0%	60% of those who receive recommendations	Primary success indicator - willingness to act on AI matches
Groups that hold at least one meeting	0%	75% of formed groups	Validates real-world activation beyond platform

Users who return for next exam cycle	0%	40% retention	Measured in subsequent semester (delayed metric)
Groups that meet 2+ times during exam cycle	0%	50% of formed groups	Indicates sustained group effectiveness

→ **REFERRAL - Users invite others organically**

Metric	Current Baseline	Target	Notes
Students who invite classmates not yet signed up	0%	25% of active users	Organic word-of-mouth indicator
Referral conversion rate	N/A	50% of invited students sign up	Quality of peer recommendations
Unsolicited social media mentions/shares	0	5+ posts in course groups	Brand awareness and satisfaction signal

→ **REVENUE (Not Applicable for Academic MVP)**

Future consideration: Institutional licensing model, premium features for cross-department matching, or freemium tier for advanced analytics

Primary Success Metrics (North Star)

Metric	Current Baseline	Target	Notes
Study Group Formation Rate	0% (manual methods unknown)	60% of active users successfully matched into study groups	Measured as: % of students with completed profiles who accept group assignment and whose group meets at least once

5. Users & Personas

5.1 Primary Target user

In scope for MVP

The primary target user is a second-year undergraduate student enrolled in academically rigorous, core courses where peer collaboration significantly impacts performance. These students experience high academic pressure, fragmented communication across sections, and difficulty forming effective study groups—particularly during exam preparation periods.

The MVP focuses on course-specific, time-bound study group formation during exam cycles, where urgency and intent to collaborate are highest.

5.2 Detailed primary Persona

Name: Arjun Reddy
 Age: 19 | Program: B.Tech Computer Science, Second Year
 Context Course: Data Structures & Algorithms

Academic Profile

Arjun is enrolled in multiple core technical courses that require both conceptual understanding and practical problem-solving. While he grasps theory well, he struggles to apply concepts under timed exam conditions. As exams approach, he seeks structured peer collaboration but lacks a reliable way to find compatible study partners across sections.

Behavioural characteristics

- Studies independently during the semester
- Shifts to intensive preparation 2–3 weeks before exams
- Uses WhatsApp and Telegram for academic communication

- Rarely engages with formal university platforms
- Hesitant to publicly request help due to social discomfort

Core Pain Points

- Study group requests scatter across WhatsApp, Telegram, and Discord—making it hard to find active groups consistently
- Forming groups requires repeated messages, follow-ups, and schedule negotiations during high-stress exam periods
- Past groups suffered from mismatched preparation levels, lack of commitment, or unclear objectives
- Reaching out to unfamiliar classmates or admitting knowledge gaps feels uncomfortable

Motivations

- Improve performance in challenging core courses through peer learning
- Maximize limited exam prep time by connecting with committed, goal-aligned peers
- Reduce the uncertainty and friction of manual group formation

Constraints

- Compressed timeline during exam cycles with multiple competing subjects
- Low trust in randomly formed groups
- High academic pressure environment

6. Scope

In-Scope

- Course-specific study group matching for a single pilot section
- AI-powered peer-complement algorithm (matching students with complementary strengths/weaknesses)
- Availability and study preference inputs (preferred times, study goals, learning style)
- Automated group formation (3-5 members per group based on compatibility)
- University SSO authentication (single sign-on via institutional credentials)
- Basic student profile creation (name, section, enrolled course)
- Exam preparation cycle focus (launch 3 weeks before mid-term exams)

Out-of-Scope

- In-app messaging or chat functionality (students use existing platforms like WhatsApp/Telegram)
- Multi-course matching (students matched across multiple enrolled courses simultaneously)
- Cross-section matching (students from different sections of the same course)
- Cross-year mentorship (seniors helping juniors)
- Study material repository or file sharing
- Flashcard creation or practice test tools

7. Requirements

7.1 Functional Requirements

Req ID	User Story / Requirement	Priority (P0/P1/P2)	Acceptance Notes
FR-1	University SSO Authentication	P0	Students must authenticate using institutional SSO credentials to verify enrollment and access the platform
FR-2	Course-Specific Profile Creation	P0	Students must create a profile for their enrolled course including: topic-level confidence rating (Weak/Average/Strong across 6-8 major topics), 2-3 topics they can help with, 2-3 topics they need help with, and primary study goal (Exam Prep/Assignment Help/Concept Clarity)
FR-3	Availability Input	P0	Students must select at least 2 preferred study time slots from predefined options (Weekday Evenings, Weekend Mornings, Weekend Afternoons, etc.)
FR-4	Peer-Complement Matching Algorithm	P0	Algorithm must match students with complementary strengths (Student A strong in Topic X pairs with Student B weak in Topic X), shared availability (minimum 2 overlapping time slots), and aligned study goals
FR-5	Compatibility Score Calculation	P0	System generates a 0-100% compatibility score for each group based on: complementary strengths (40%), availability overlap (30%), goal alignment (20%), balanced group composition (10%)
FR-6	Single Group Recommendation	P0	Each student receives one curated study group recommendation (3-5 members) with compatibility score and brief rationale explaining the match (e.g., "Priya is strong in graphs where you need help")
FR-7	Accept/Decline Group Assignment	P0	Students must explicitly accept or decline their recommended group; no automatic enrollment
FR-8	Group Activation & Contact Exchange	P0	Study group activates when minimum 3 members accept the recommendation; upon activation, all members receive contact details (email/phone) for external coordination
FR-9	No-Match Handling	P1	If algorithm cannot find compatible matches for a student (insufficient user pool or no complementary profiles), system notifies them and suggests trying again when more students join
FR-10	Post-Formation Satisfaction Survey	P1	Students receive a brief survey 1 week after group formation: "Did your group meet?", "Satisfaction rating (1-5)", "Would you recommend this to classmates?"
FR-11	Usage Analytics Tracking	P0	System logs key events for success measurement: sign-ups, profile completions, match recommendations sent, group acceptances, group formations

FR-12	Admin Dashboard	P1	Admin can view real-time metrics (total sign-ups, profile completion rate, groups formed, acceptance rate) and manually trigger matching algorithm after sufficient profiles are completed
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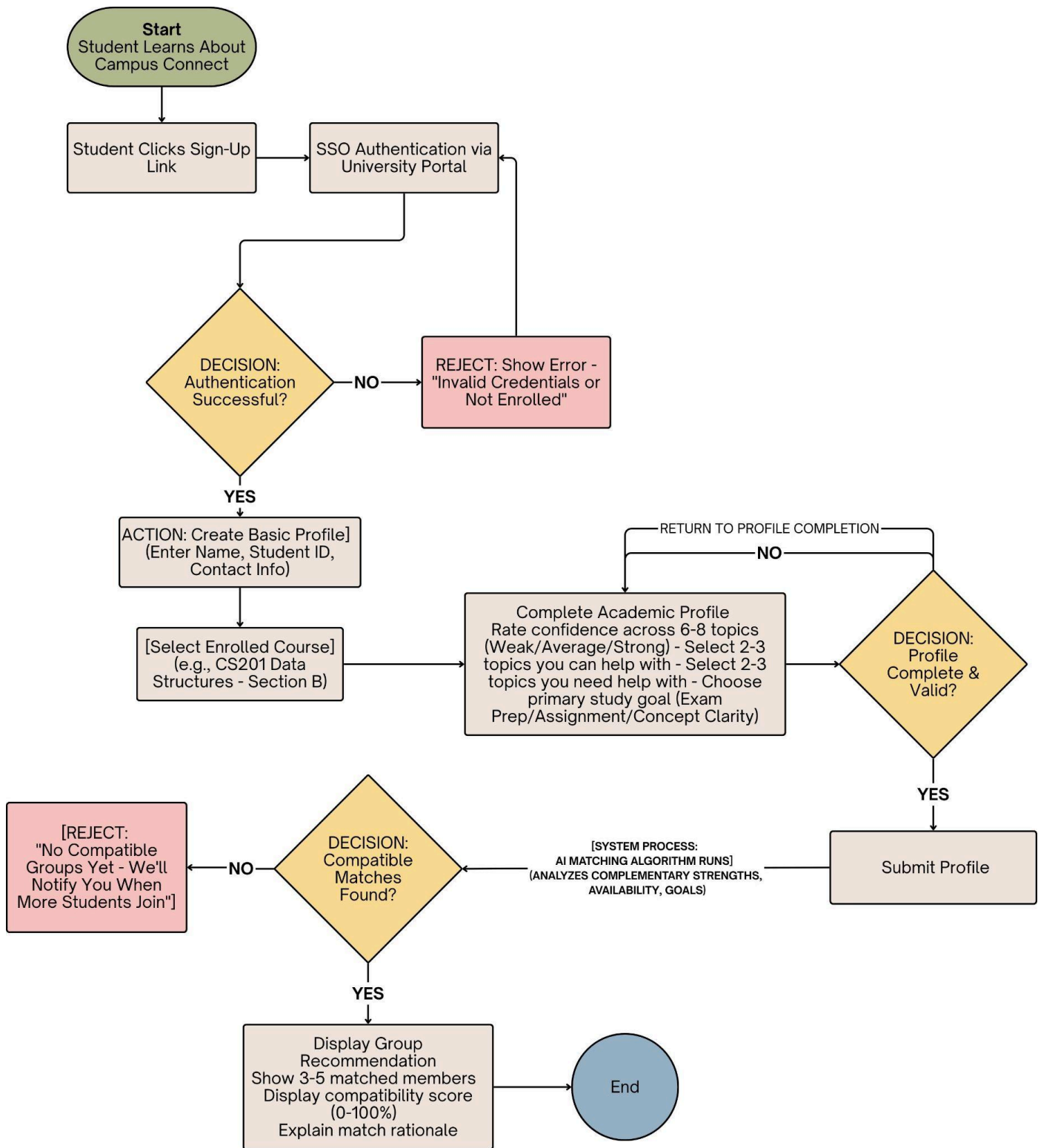
Priority meaning:

- P0 - Must-have for release
- P1 - Should-have
- P2 - Nice-to-have / later

7.2 Non-Functional Requirements

Area	Requirement
Performance	Response Time, System Capacity
Security	Data Encryption, Authentication, Data Access Control, Consent Management
Usability	Mobile Responsiveness, Onboarding Efficiency, Accessibility
Compliance	Uptime, Data Backup, University Policies
Scalability	User Load,
Logging/Audits	Code Documentation

8. User flow diagram



9. Risks & Mitigation

Risk	Impact	Probability	Mitigation
Low adoption during pilot	Medium	High	Launch 3 weeks before exams (high urgency); incentivize early sign-ups with professor endorsement
Inaccurate self-assessment	Medium	Medium	Include validation examples; refine algorithm based on feedback; accept some noise in MVP
Privacy concerns prevent sign-ups	Low	High	Clear consent messaging; highlight data-sharing only with matched members; no public profiles
Matched groups don't actually meet	Medium	High	Follow up with 1-week survey; if >40% don't meet, indicates fundamental problem-solution fit failure
Algorithm perceived as ineffective	Medium	High	Provide transparent match rationale; collect satisfaction data; be ready to pivot if <50% acceptance rate
Technical failures during peak usage	Low	Medium	Load test before launch; have manual backup process for critical issues

10. Release Plan

10.1 Rollout

Rollout Type: Phased Pilot

Target Audience: CS201 Data Structures - Section B
(Second Year, ~60 students)

Timeline: 3-week deployment aligned with mid-term exam preparation cycle

Phase Breakdown:

Phase	Timeline	Activities	Success Criteria
Pre-Launch	Week -1	Configure course topics; test with 10 beta users; finalize SSO integration	All functional requirements tested; zero critical bugs
Launch	Week 1 (Day 1-7)	Announce via email/class; professor endorsement; onboarding support available	35+ sign-ups; 25+ completed profiles
Matching	Week 2 (Day 8-10)	Run AI algorithm; send recommendations; support acceptance process	15+ groups formed; 60%+ acceptance rate

Active Use	Week 2-3 (Days 11-21)	Monitor group activity; provide support; collect early feedback	75%+ of groups meet at least once
Post-Exam	Week 4+	Send satisfaction surveys; analyze data; compile learnings	70%+ satisfaction; clear decision on scale/pivot

11. OPEN QUESTIONS

11.1 QA Scenarios

- **Unmatched Users & Group Size Flexibility:** How do we handle students who don't fit into compatible groups? Should we lower matching standards, allow 2-person pairs instead of 3-5 member groups, or keep them unmatched until more users join?
- **Re-matching Policy:** If a matched group dissolves or doesn't work out in the first week, should we offer one-time re-matching? What's the policy to prevent users from endlessly cycling through groups?
- **Conflicting Self-Assessments:** How do we handle contradictions when students rate themselves "Strong" in a topic but also list it under "need help with"? Should we flag for correction or use algorithm logic to resolve?
- **Group Activity Verification:** Beyond self-reported surveys, do we need to verify that groups actually met and studied together? Or is mutual acceptance and contact exchange sufficient for MVP validation?
- **Multi-Course Expansion Strategy:** When scaling beyond single course, should one student be allowed in multiple study groups simultaneously (one per course)? What's the realistic time commitment threshold to prevent overload?
- **Algorithm Validation Method:** Should we implement a control group (20% random matching) to A/B test AI effectiveness, or rely on post-pilot surveys comparing to past manual group formation experiences?